OOPJ DAILY ASSIGNMENT 7 (18/12/25)

Name – Nishant Nahar

Roll Number - 241551078

1. Class Instance
   a. Without This

```java
class Student {
    int rollno;
    String name;
    float fee;
    Student(int rollno, String name, float fee) {
        rollno = rollno;
        name = name;
        fee = fee;
    }
    void display() {
        System.out.println(rollno + " " + name + " " + fee);
    }}
public class TestThis1 {
    public static void main(String args[]) {
        System.out.println("Nishant Nahar --241551078");
        Student s1 = new Student(111, "ankit", 5000f);
        Student s2 = new Student(112, "sumit", 6000f);
        s1.display();
        s2.display();
    }}
```

OUTPUT

```
PS B:\java_theory\assignment_7> java .\TestThis1.java
Nishant Nahar --241551078
0 null 0.0
0 null 0.0
PS B:\java_theory\assignment_7>
```

   b. With This

```java
class Student {
    int rollno;
    String name;
    float fee;
    Student(int rollno, String name, float fee) {
        this.rollno = rollno;
        this.name = name;
        this.fee = fee;
    }
    void display() {
        System.out.println(rollno + " " + name + " " + fee);
```

```
    }}
public class TestThis {
    public static void main(String args[]) {
        System.out.println("Nishant Nahar -- 241551078");
        Student s1 = new Student(111, "ankit", 5000f);
        Student s2 = new Student(112, "sumit", 6000f);
        s1.display();
        s2.display();
    }}
```

OUTPUT

```
PS B:\java_theory\assignment_7> java .\TestThis.java
Nishant Nahar -- 241551078
111 ankit 5000.0
112 sumit 6000.0
PS B:\java_theory\assignment_7>
```

2. Current Class Method
    a. With This

```
class A {
    void m() {
        System.out.println("hello m");
    }
    void n() {
        System.out.println("hello n");
        this.m();
    }}
class TestThis {
    public static void main(String args[]) {
        System.out.println("Nishant Nahar -- 241551078");
        A a = new A();
        a.n();
    }}
```

OUTPUT

```
PS B:\java_theory\assignment_7> java TestThis
Nishant Nahar -- 241551078
hello n
hello m
```

    b. Without This

```
class A {
    void m() {
        System.out.println("hello m");
    }

    void n() {
        System.out.println("hello n");
        m();
    }}
```

```java
class TestThis {
    public static void main(String args[]) {
        System.out.println("Nishant Nahar -- 241551078");
        A a = new A();
        a.n();
    }}
```

OUTPUT

```
PS B:\java_theory\assignment_7> java TestThis
Nishant Nahar -- 241551078
hello n
hello m
```

3. Invoke Class Constructor
    a. Default called

```java
class A {
    A() {
        System.out.println("hello a");
    }
    A(int x) {
        this();
        System.out.println(x);
    }}
public class TestThis {
    public static void main(String args[]) {
        A a = new A(10);
    }}
```

OUTPUT

```
PS B:\java_theory\assignment_7> java .\TestThis.java
hello a
10
```

    b. Parameterized Called

```java
class A {
    A() {
        this(5);
        System.out.println("hello a");
    }
    A(int x) {
        System.out.println(x);
    }
}
public class TestThis {
    public static void main(String args[]) {
        A a = new A();

    }
}
```

OUTPUT

```
PS B:\java_theory\assignment_7> java .\TestThis.java
5
hello a
```

4. Constructor Chaining

```
class Student {
    int rollno;
    String name, course;
    float fee;
    Student(int rollno, String name, String course) {
        this.rollno = rollno;
        this.name = name;
        this.course = course;
    }
    Student(int rollno, String name, String course, float fee) {
        this(rollno, name, course);
        this.fee = fee;
    }
    void display() {
        System.out.println(rollno + " " + name + "" + course + " " +
fee);
    }
}
public class TestThis {
    public static void main(String args[]) {
        Student s1 = new Student(111, "ankit", "java");
        Student s2 = new Student(112, "sumit", "java", 6000f);
        s1.display();
        s2.display();
    }}
```

OUTPUT

```
PS B:\java_theory\assignment_7> java .\TestThis.java
111 ankitjava 0.0
112 sumitjava 6000.0
```

5. Check : call to this() must be the first statement in constructor

```
public class S3 {
    int rollno;
    String name, course;
    float fee;
    S3(int rollno, String name, String course) {
        this.rollno = rollno;
        this.name = name;
        this.course = course;
    }
    S3(int rollno, String name, String course, float fee) {
```

```
        this.fee = fee;
        this(rollno, name, course);// C.T.Error
    }}
```

OUTPUT

```
PS C:\Users\KIIT\Desktop\java\class\src> javac S3.java
S3.java:12: error: flexible constructors is a preview feature and is disabled by default.
            this.fee = fee;
            ^
  (use --enable-preview to enable flexible constructors)
1 error
```

6. This: to pass as an argument in the method

```
public class S2 {
    void m(S2 obj) {
        System.out.println("method is invoked");
    }
    void p() {
        m(this);
    }
    public static void main(String args[]) {
        System.out.println("Nishant Nahar -- 241551078");
        S2 s1 = new S2();
        s1.p();
        ;
    }
}
```

OUTPUT

```
PS B:\java_theory\assignment_7> java .\S2.java
Nishant Nahar -- 241551078
method is invoked
```

7. This: to pass as argument in the constructor call

```
public class A3 {
    int data = 10;
    A3() {
        A1 b = new A1(this);
        b.display();
    }
    public static void main(String args[]) {
        System.out.println("Nishant Nahar -- 241551078");
        A3 a = new A3();
    }
}
public class A1 {
    A3 obj;
    A1(A3 obj) {
        this.obj = obj;
    }
```

```
    void display() {
        System.out.println(obj.data);// using data member of A class
    }
}
```

OUTPUT

```
PS B:\java_theory\assignment_7> java .\S2.java
Nishant Nahar -- 241551078
10
```

8. This keyword can be used to return current class instance

```
public class A4 {
    A4 getA() {
        return this;
    }
    void msg() {
        System.out.println("Hello java");
    }
public static void main(String[] args){
System.out.println("Nishant Nahar -- 241551078");
new A4().getA().msg();
}
```

OUTPUT

```
PS B:\java_theory\assignment_7> java .\A4.java
Nishant Nahar -- 241551078
Hello java
```

9. Proving This Keyword

```
public class A5 {
    void m() {
        System.out.println(this);// prints same reference ID
    }
    public static void main(String args[]) {
        System.out.println("Nishant Nahar -- 241551078");
        A5 obj = new A5();
        System.out.println(obj);// prints the reference ID
        obj.m();
    }
}
```

OUTPUT

```
PS B:\java_theory\assignment_7> java .\A5.java
Nishant Nahar -- 241551078
A5@57f23557
A5@57f23557
```